รายงานผลLab1

กลุ่มเรียน (Section): Sec-2

1. สมาชิกกลุ่มและหน้าที่รับผิดชอบ

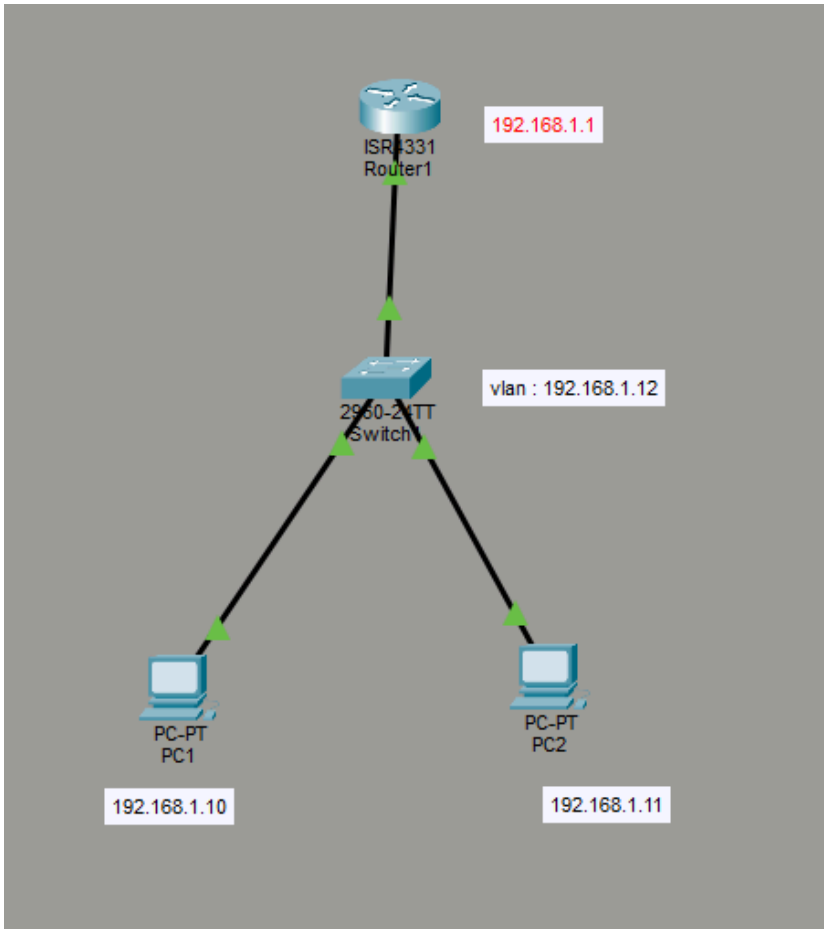
ธนธัช วรรณไทย 673380403-2 รับผิดชอบ ตอบคำถามและเอกสาร

อนัตตา โยคาพจร 673380429-4 รับผิดชอบ worksheet 1

จีรภัทร แก้วดี 673380577-9 รับผิดชอบตั้งค่าอุปกรณ์และเครือข่าย
ธีธัช ลิ้มประยูรวงศ์ 673380408-2 รับผิดชอบ ตอบคำถาม
ยุทธนา เหล่าวิสัย 673380422-8 รับผิดชอบ worksheet 3

2. อุปกรณ์และการเชื่อมต่อ

2.1แผนภาพเครือข่าย



3.2 Addressing Table

Device (อุปกรณ์)	Interface	IP Address	Subnet Mask	Default Gateway
R1 (Router)	GigabitEthernet0/0/0	192.168.1.1	255.255.255.0	-
S1 (Switch)	VLAN 1	192.168.1.12	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1

3.การตั้งค่าอุปกรณ์

ได้ทำการตั้งค่าอุปกรณ์ตามขั้นตอนดังนี้:

1. **Router R1:** เปลี่ยนชื่อ Hostname, และกำหนด IP Address G0/0/0 เป็น 192.168.1.1/24
2. **Switch S1:** กำหนด Management IP Address บน VLAN 1 เป็น 192.168.1.12/24
3. **PC1 & PC2:** กำหนด Static IP และ Default Gateway ให้ชี้ไปที่ R1

```
R1#sh ip interface bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

4: การตรวจสอบการเชื่อมต่อ

ทำการทดสอบการเชื่อมต่อโดยใช้คำสั่ง Ping จากเครื่อง PC1 ไปยังอุปกรณ์ต่างๆ ในเครือข่าย

LAB DELIVERABLES

Worksheet 1: Configuration Verification

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	show ip int brief	G0/0 UP/UP	G0/0/0 UP/UP	Pass
S1 VLAN Config	show vlan brief	VLAN 1 Active	VLAN 1 Active	Pass
PC1 Connectivity	ping 192.168.1.11	Success (4 replies)	Success 4 replies	Pass
ARP Resolution	arp -a on PC1	2+ entries	Show 2 entries router and other pc	Pass
MAC Learning	show mac address-table on S1	2 dynamic entries	2 dynamic entries	Pass

5: การวิเคราะห์ Protocol

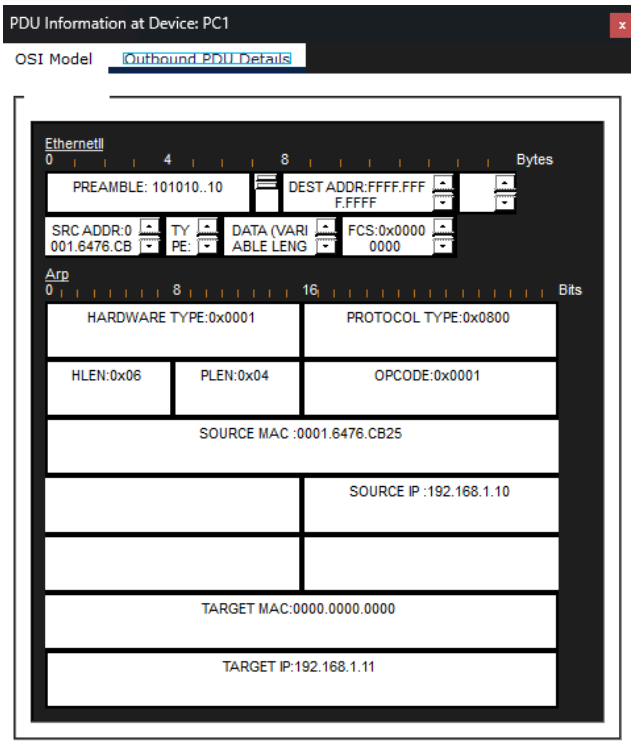
เปลี่ยนโหมดเป็น **Simulation Mode** เพื่อดูรายละเอียด

บันทึกการวิเคราะห์

Worksheet 2: Protocol Analysis

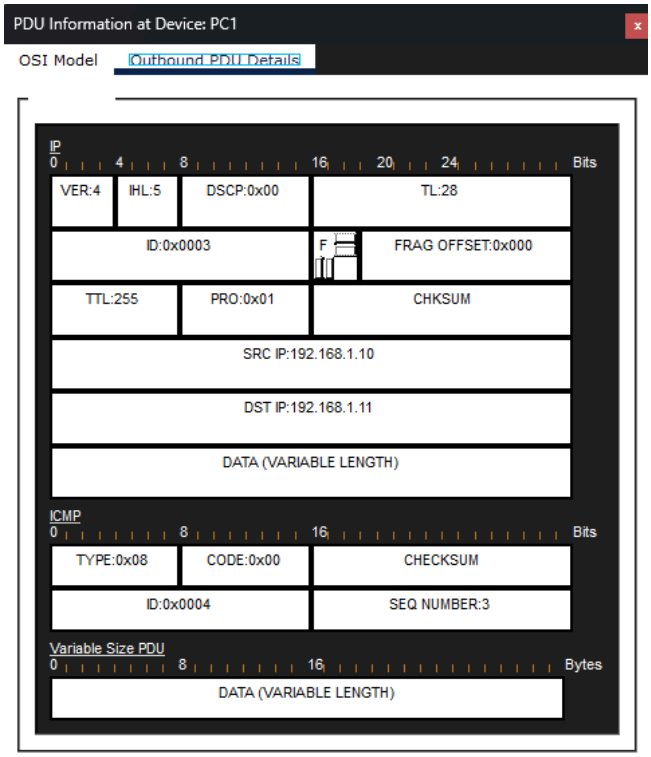
1. **ARP Request Frame:**

    - Source MAC: 0001.63E8.2001
    - Destination MAC: FFFF.FFFF.FFFF
    - ARP Opcode: 1



2. **ICMP Echo Packet:**

- Time to Live (TTL): 255
- Protocol Field: 0x01 (Value = 1)



3. **Encapsulation Chain:** Data → Ethernet II Header → IP Header → Ethernet II Header → Bits

ส่วนที่ 4: การแก้ปัญหาเครือข่าย (Troubleshooting Scenario)

ผู้รับผิดชอบ:

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	Request timed out ไม่สามารถ ping ไปหาอีกได้ และ ไม่พบ Entries	ping, arp -a	Subnet mask ที่ผิดหรือต่างกัน	แก้ subnet mask ให้ถูก
Interface Shutdown	Gigabit 0/0/0 down down	show ip int brief	Router ถูกปิด	ไปเปิด router โดยใช้ no shutdown

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Incorrect Gateway	Request timed out ไม่สามารถเข้าถึงได้	ipconfig, tracert	การตั้ง gateway ผิด	ตั้งgateway ให้ถูก

Router (R1):

- ☑ `show running-config`

```
R1#show running-config
Building configuration...

Current configuration : 720 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
```

```
interface GigabitEthernet0/0/0
 description LINK_TO_SWITCH_S1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 Null0
!
ip flow-export version 9
```

- ☑ show ip interface brief

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

- ☑ show arp

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -    0001.63E8.2001  ARPA   GigabitEthernet0/0/0
```

- ☑ show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
S*   0.0.0.0/0 is directly connected, Null0
```

Switch (S1):

- ☑ show running-config

```
S1#show running-config
Building configuration...

Current configuration : 1171 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode access
!
interface FastEthernet0/2
 switchport mode access
!
interface FastEthernet0/3
!
interface FastEthernet0/4
```

- [x] show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

- [x] show mac address-table

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.63e8.2001    DYNAMIC     Gig0/1
```

- [x] show interfaces status

```
S1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        connected    1          a-full  a-100 10/100BaseTX
Fa0/3                        notconnect   1          auto    auto  10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
Fa0/22                       notconnect   1          auto    auto  10/100BaseTX
Fa0/23                       notconnect   1          auto    auto  10/100BaseTX
Fa0/24                       notconnect   1          auto    auto  10/100BaseTX
Gig0/1                       connected    1          a-full  a-100 10/100/1000BaseTX
Gig0/2                       notconnect   1          auto    auto  10/100/1000BaseTX
```

End Devices:

- ☐ `ipconfig /all` (or equivalent)

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 0001.6476.CB25
   Link-local IPv6 Address.........: FE80::201:64FF:FE76:CB25
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-0D-82-43-A4-00-01-64-76-CB-25
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address.................: 000A.F3D7.C51E
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-0D-82-43-A4-00-01-64-76-CB-25
   DNS Servers.....................: ::
                                     0.0.0.0
```

- [x] `arp -a`

- [x] `ping` to all devices

Ping router

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping pc2

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping switch

```
C:\>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time<1ms TTL=255
Reply from 192.168.1.12: bytes=32 time<1ms TTL=255
Reply from 192.168.1.12: bytes=32 time<1ms TTL=255
Reply from 192.168.1.12: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

1. **When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?**
   : ARP

2. **What would happen if you changed the subnet mask on PC1 to 255.255.255.128?**
   : PC1 อาจ ping PC2 ไม่ได้

3. **Why does the switch not need an IP address for basic frame forwarding?**
   : เพราะ Switch ทำงานที่ OSI Layer 2

4. **Which OSI layers are involved when R1 forwards a packet between interfaces?**

   : Layer 1, Layer 2, และ Layer 3

5. **What is the purpose of the TTL field in the IP header you observed?**

   : ป้องกัน packet loop ในเครือข่าย